

LIFE LINCO - SPEC 667MG SOLUBLE POWDER

FOR ANIMAL USE ONLY

Brand or Product Name :

LIFE LINCO-SPEC 667MG SOLUBLE POWDER

Name and Strength of Active Substance(s)

Each gram of LIFE LINCO-SPEC 667MG SOLUBLE POWDER contains Lincomycin 222 mg (equivalent to Lincomycin Hydrochloride 251.74 mg) and Spectinomycin 445 mg (equivalent to Spectinomycin Sulfate Tetrahydrate 672.81 mg).

Product Description

LIFE LINCO-SPEC 667MG SOLUBLE POWDER is a white or off-white powder for use in drinking water.

Target Species

Chickens and pigs.

Pharmacology Properties

LIFE LINCO-SPEC 667MG SOLUBLE POWDER contains the antibiotics lincomycin and spectinomycin, having a complementary spectrum of activity.

Lincomycin

Lincomycin is a lincosamide antibiotic derived from *Streptomyces lincolnensis* which inhibits protein synthesis. Lincomycin binds to the 50S sub-unit of the bacterial ribosome close to the peptidyl transfer centre and interferes with the peptide chain elongation process by inhibiting the formation of the 50S sub-unit and by stimulating ribosomal peptidyl-tRNA dissociation. Lincomycin is active against gram-positive bacteria, some anaerobic gram-negative bacteria (such as *Brachyspira hyodysenteriae*) and mycoplasmas. It has little or no action against gram-negative bacteria such as *Escherichia coli*. While the lincosamides are generally considered to be bacteriostatic agents, the activity is depending on the sensitivity of the organism and concentration of the antibiotic. Lincomycin may be either bactericidal or bacteriostatic. Resistance to lincomycin is frequently conferred by plasmid-borne factors (erm genes) coding for methylases modifying the ribosomal binding site and frequently leading to cross-resistance to other antimicrobials of the MLS_B group. However, the most prevalent mechanism in *B. hyodysenteriae* and mycoplasmas is the alteration of the binding site through mutational events (chromosomal resistance). Lincomycin resistance mediated by efflux pumps, or by inactivating enzymes, has also been described. There is often complete cross-resistance between lincomycin and clindamycin. Resistance to lincomycin can readily develop in *B. hyodysenteriae* and most of the isolates studied show decreased susceptibility *in vitro*.

Spectinomycin

Spectinomycin is an aminocyclitol antibiotic derived from *Streptomyces spectabilis*, it has bacteriostatic activity and is active against *Mycoplasma* spp. and against some gram-negative bacteria such as *E. coli*. The mechanism by which spectinomycin administered orally acts on pathogens at the systemic level despite a poor absorption is not fully elucidated, and might rely partly on indirect effects on the gut flora, or on effects of metabolite(s). Chromosomal one-step mutation to high-level spectinomycin resistance develops in many enteric bacteria (such as *E. coli*). Plasmid-mediated resistance is less common. Strains with chromosomal resistance do not show cross-resistance with aminoglycosides. In *E. coli* and *Salmonella* spp. the MIC distribution appears to be bimodal, with a significant number of strains showing high values; this could partly correspond to natural (intrinsic) resistance. *In vitro* studies as well as clinical efficacy data show that the lincomycin-spectinomycin combination is active against *Lawsonia intracellularis*. Due to technical constraints the susceptibility of *Lawsonia intracellularis* is difficult to test *in vitro*, and data about the resistance status in that species are lacking.

Pharmacokinetics Properties

In pigs, lincomycin is rapidly absorbed following oral administration. Lincomycin pharmacokinetic studies in pigs show that lincomycin is bioavailable when given intravenously, intramuscularly or orally. Studies performed in various animal species have demonstrated that spectinomycin undergoes limited absorption from the intestine after oral administration. Spectinomycin exhibits little tendency to protein binding and is poorly liposoluble.

Indications

Pigs

For the treatment and prevention of dysentery caused by *Brachyspira hyodysenteriae* and for the treatment and prevention of porcine proliferative enteropathy (ileitis) caused by *Lawsonia intracellularis*, and associated enteric pathogens (*Escherichia coli*). The presence of the disease in the group must be established before the product is used.

Chickens

For the treatment and prevention of chronic respiratory disease (CRD) caused by *Mycoplasma gallisepticum* and *Escherichia coli*, and associated with a low mortality rate. The presence of the disease in the flock must be established before the product is used.

Route of Administration

For use in drinking water.

Recommended Dosage

The recommended dosage rates are:

Pigs:

15 mg LIFE LINCO-SPEC 667MG / kg bw / day for 7 days. This dose is correspond to 10 mg total antibiotic activity/ kg bw / day.

Amount of water consumed by pig (ml) / kg bw / day	150 g of LIFE LINCO-SPEC 667MG should be diluted in...	This dose corresponds to ...
100 ml	1,000 L of drinking water	15 mg LIFE LINCO-SPEC 667MG / kg bw / day or 10mg total antibiotic activity/ kg bw / day
150 ml	1,500 L of drinking water	
200 ml	2,000 L of drinking water	
250 ml	2,500 L of drinking water	

In pigs 150 g of the veterinary medicinal product corresponds to the dose for 10,000 kg of body weight per day.

To determine the volume of dilution (in litres of drinking water) required for 150 g of the veterinary medicinal product, use the following formula :

Volume (L) for 150 g of LIFE LINCO-SPEC 667MG	=	$\frac{10,000 \times [\text{daily water consumption per animal (L)}]}{\text{average body weight of one pig (kg)}}$
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Chickens:

75 mg LIFE LINCO-SPEC 667MG / kg bw/day for 7 days. This dose is correspond to 50 mg total antibiotic activity/ kg bw / day.

Amount of water consumed by chicken (ml) / kg bw / day	150 g of LIFE LINCO-SPEC 667MG should be diluted in...	This dose corresponds to ...
50 ml	100 L of drinking water	75 mg LIFE LNCO-SPEC 667MG / kg bw / day or 50mg total antibiotic activity / kg bw / day
100 ml	200 L of drinking water	
150 ml	300 L of drinking water	
200 ml	400 L of drinking water	
250 ml	500 L of drinking water	

In chickens, 150 g of the veterinary medicinal product corresponds to the dose for 2,000 kg of body weight per day.

To determine the volume of dilution (in litres of drinking water) for 150 g of the veterinary medicinal product, use the following formula:

Volume (L) for 150 g of LIFE LINCO-SPEC 667MG	=	$\frac{2,000 \times [\text{daily water consumption per bird (L)}]}{\text{average body weight of one bird (kg)}}$
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Direction of Use

Treatment should be initiated as soon as first clinical signs occur. For the preparation of drinking water, the incorporation rate of the veterinary medicinal product in water will depend on the body weight of the animals and their actual daily intake of water. To ensure a correct dosage and avoid underdosing, mean body weights in the group of animals and daily water consumption should be determined as accurately as possible. The medicated drinking water should be the sole source of drinking water for the treatment duration. Any medicated water which is not consumed within 24 hours should be discarded. In case of disease accompanied with significant decrease in water intake, parenteral treatment may have to be initiated.

Contraindications

Do not use in case of hypersensitivity to the active substances or any of the excipients. Do not use in case of hepatic dysfunction. Do not allow rabbits, rodents (e.g. chinchillas, hamsters, guinea pigs), horses or ruminants to access to water or feeds containing lincomycin. Ingestion by these species may result in severe gastrointestinal effects. Do not use in laying hens

Warnings

In *E. coli*, a significant part of the strains show high MIC values (minimum inhibitory concentrations) against the lincomycin-spectinomycin combination and may be clinically resistant, although no breakpoint is defined. Due to technical constraints the susceptibility of *L. intracellularis* is difficult to test *in vitro*, and data about the lincomycin-spectinomycin resistance status in that species are lacking.

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Precautions

i. Special precautions for use in animals

Lincomycin resistance is widespread in *B. hyodysenteriae* and may lead to clinical treatment failure. It is sound clinical practice to base treatment on susceptibility testing of the bacteria isolated from the animal. If this is not possible, therapy should be based on local (regional, farm level) epidemiological information about susceptibility of target bacteria. Use of the veterinary medicinal product deviating from the instructions in this pamphlet may increase the risk of development and selection of resistant bacteria and decrease the effectiveness of treatment with macrolides due to the potential for cross-resistance. The oral use of preparations containing lincomycin is only indicated in swine and chickens. Do not leave access to the medicated water for other animals. Lincomycin may lead to severe gastrointestinal disturbances in other animal species. The repeated or prolonged use should be avoided, by improving the farm management and disinfection practices. Diagnosis should be reconsidered if improvement is not seen after 5 days. Sick animals have a reduced appetite and an altered drinking pattern, and severely affected animals may therefore require parenteral treatment.

This powder is for use in drinking water only and should be dissolved before use.

ii. Special precautions to be taken by the person administering the veterinary medicinal product to animals

People with known hypersensitivity to lincomycin, spectinomycin or soybean millfeed should avoid contact with the veterinary medicinal product. Care should be taken not to raise and inhale any dust. Contact with skin and eyes should be avoided. Personal protective equipment consisting of approved dust masks, gloves and safety glasses should be worn when handling and mixing the product. Wash hands and any exposed skin with soap and water immediately after use. If symptoms such as skin rash or persistent eye irritation appear after exposure, seek medical advice immediately and show the package leaflet or label to the physician.

Drug Interactions

In general mixture with other medicines should be avoided. The combination of lincosamides and macrolides is antagonistic, due to competitive binding to their target sites. Combination with anaesthetics may lead to possible neuromuscular blocking. Do not administer with kaolin or pectine as they impair lincomycin absorption. If co-administration is mandatory, respect a delay of two hours between intakes.

Pregnancy and Lactation

Pigs : The safety of the veterinary medicinal product has not been established during pregnancy and lactation. Laboratory studies in dogs and rats have not produced any evidence of reproductive, foetotoxic or teratogenic effects for lincomycin or spectinomycin. Lincomycin is excreted in milk. Use only according to benefit-risk assessment by the responsible veterinarian.

Chickens : Do not use in birds in lay.

Adverse Effects

Cases of diarrhoea or soft faeces and/or perianal region inflammation have been encountered in healthy pigs at the start of treatment. The symptoms disappeared within 5 to 8 days without interruption of the treatment. Rare cases of irritability/excitation, skin rash/pruritus were also observed. Allergic/hypersensitive reactions are rare but can occur and require stopping treatment with the veterinary medicinal product. A symptomatic treatment must be implemented.

Overdose and Treatment

In the event of overdose in pigs, a change in the consistency of the faeces (soft faeces and/or diarrhoea) may be observed. In chickens treated at several times the recommended dose, enlargement of the caecum and abnormal caecum content was observed. In case of accidental overdose, the treatment should be interrupted and restarted at the recommended dose.

Withdrawal Period(s)

Pigs:

Meat and offal: Zero days.

Chickens:

Meat and offal: 5 days.

Not authorised for use in birds producing eggs for human consumption, including replacement chicks which are intended to produce eggs for human consumption.

Animals must not be slaughtered for human consumption during treatment.

Shelf-life

Shelf life as packaged for sale: 3 year

Shelf life after first opening : 24 hours

Shelf life after reconstitution : 24 hours

Maximum Residual Limit (MRL)

Lincomycin

Pig : Muscle 200 µg/kg, Liver 500 µg/kg, Kidney 1500 µg/kg, Fat 100 µg/kg, Skin with adhering fat 300 µg/kg

Chicken : Muscle =200 µg/kg, Liver = 500 µg/kg, Kidney = 500 µg/kg, Fat = 100 µg/kg, Skin with adhering fat = 300 µg/kg [Ref. CODEX ALIMENTARIUS : CAC/MRL 2-2014]

Spectinomycin

Chicken and pig : Muscle = 500 µg/kg, Fat = 2000 µg/kg, Liver = 2000 µg/kg, Kidney = 5000 µg/kg, Eggs = 2000 µg/kg [Ref. CODEX ALIMENTARIUS : CAC/MRL 2-2014]

Storage Conditions

Store in a dry place below 30°C. Keep out of reach of children. Jauhi ubat daripada kanak-kanak. Any opened sachet or unused reconstituted water must be discarded after 24 hours.

Disposal of Containers

Emptied aluminum sachet must be safely disposed in trash according to local government regulations.

Dosage Forms and Packaging Available

150g, 300g, 600g, 900g in aluminum sachet.

Name and Address of Manufacturer and Product Registration Holder

Life Biopharma Sdn. Bhd. (Company No. 546163-D)

No. 250-251, Jalan Industri Galla 13, Kawasan Perindustrian Galla, 70200 Seremban, Negeri Sembilan, Malaysia.

Date of Revision of Package Insert

2016-10-18